

Problem 25.10

Show that

$$\frac{d}{dt}a_{\overline{t}|} = \frac{v^t}{\overline{s}_{\overline{1}|}}.$$

Solution

Recall that

$$a_{\overline{t}|} = \frac{1 - v^t}{i}.$$

Since i is constant with respect to t ,

$$\frac{d}{dt}a_{\overline{t}|} = \frac{d}{dt} \left(\frac{1 - v^t}{i} \right) = -\frac{1}{i} \frac{d}{dt}v^t.$$

Now write

$$v^t = (1 + i)^{-t}.$$

Let

$$y = (1 + i)^{-t}.$$

Then

$$\ln y = -t \ln(1 + i).$$

Exponentiating,

$$y = e^{-t \ln(1+i)}.$$

Therefore,

$$\frac{d}{dt}v^t = \frac{d}{dt}e^{-t \ln(1+i)} = -\ln(1 + i)e^{-t \ln(1+i)}.$$

Since

$$\delta = \ln(1 + i) \quad \text{and} \quad e^{-t \ln(1+i)} = v^t,$$

we have

$$\frac{d}{dt}v^t = -\delta v^t.$$

Thus,

$$\frac{d}{dt}a_{\overline{t}|} = -\frac{1}{i}(-\delta v^t) = \frac{\delta v^t}{i}.$$

Now note that a continuous annuity-certain of length 1 has accumulated value

$$\overline{s}_{\overline{1}|} = \frac{e^\delta - 1}{\delta}.$$

Since $e^\delta = 1 + i$,

$$\overline{s}_{\overline{1}|} = \frac{i}{\delta}.$$

Therefore,

$$\frac{1}{\bar{s}_{\mathbb{I}}] } = \frac{\delta}{i}.$$

Substitute this into the derivative expression:

$$\frac{\delta v^t}{i} = \frac{v^t}{\bar{s}_{\mathbb{I}}]}.$$

Thus,

$$\boxed{\frac{d}{dt} a_{\bar{i}}] = \frac{v^t}{\bar{s}_{\mathbb{I}}]}.$$